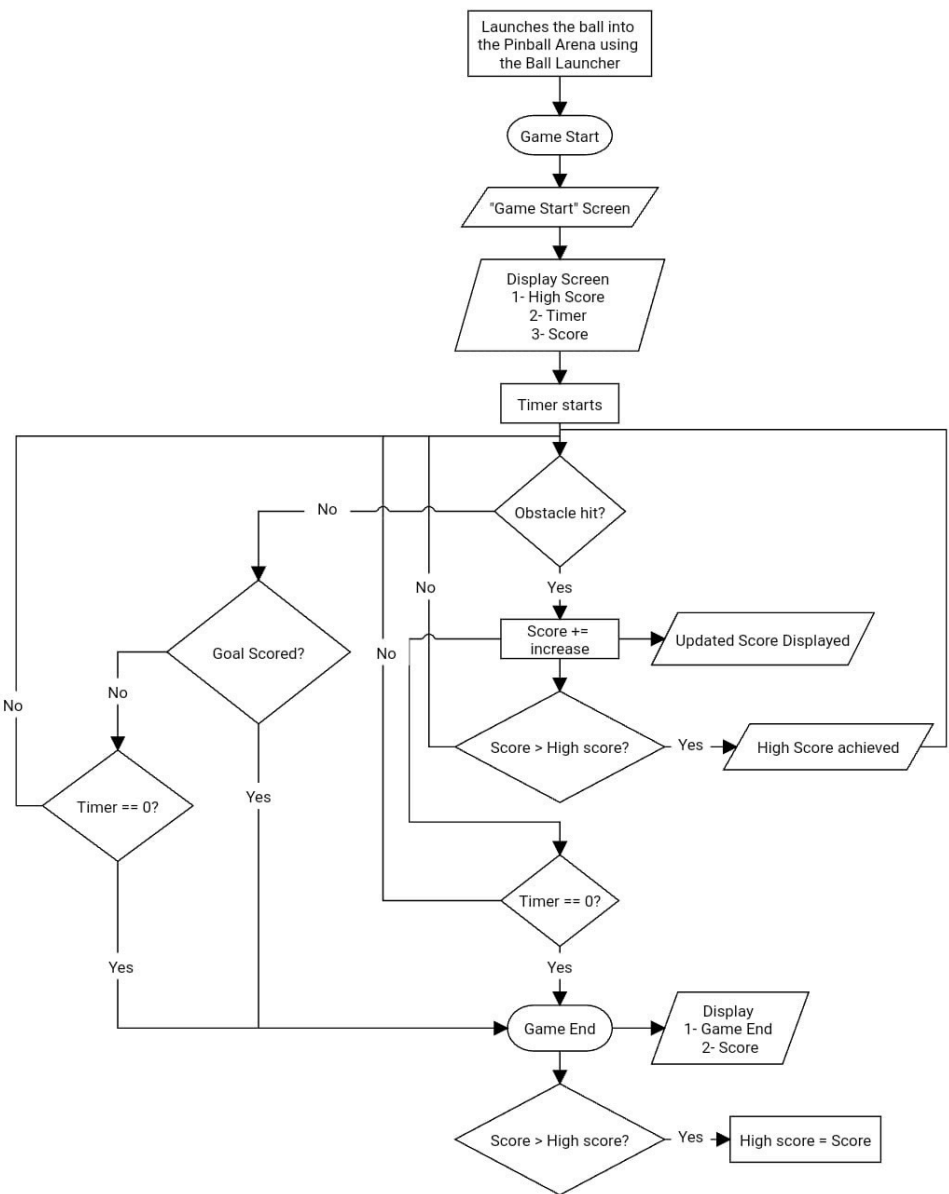




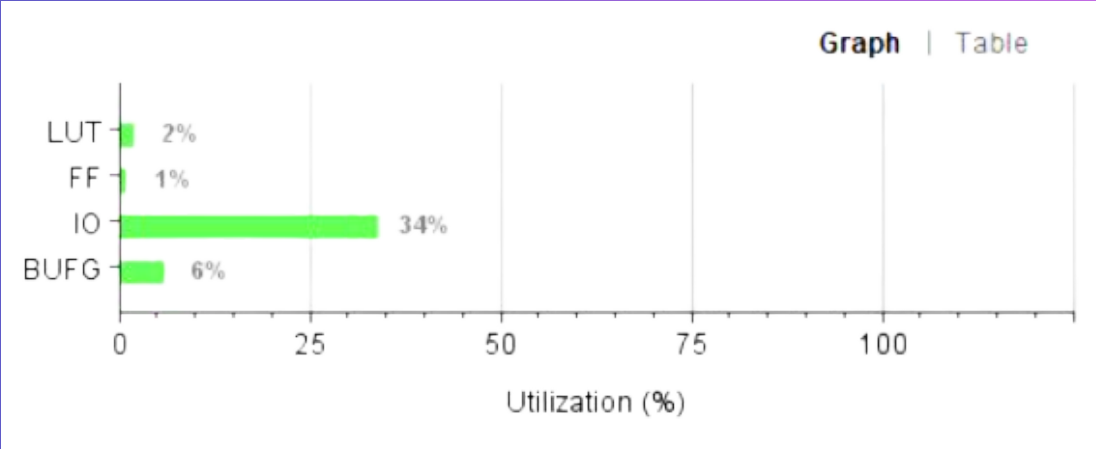
COSMIC PINBALL MACHINE 3000

MUHAMMAD HUNAIN RAZA
ABEER MASROOR
NIGARISH NAVEED
IMSHA REHMAN

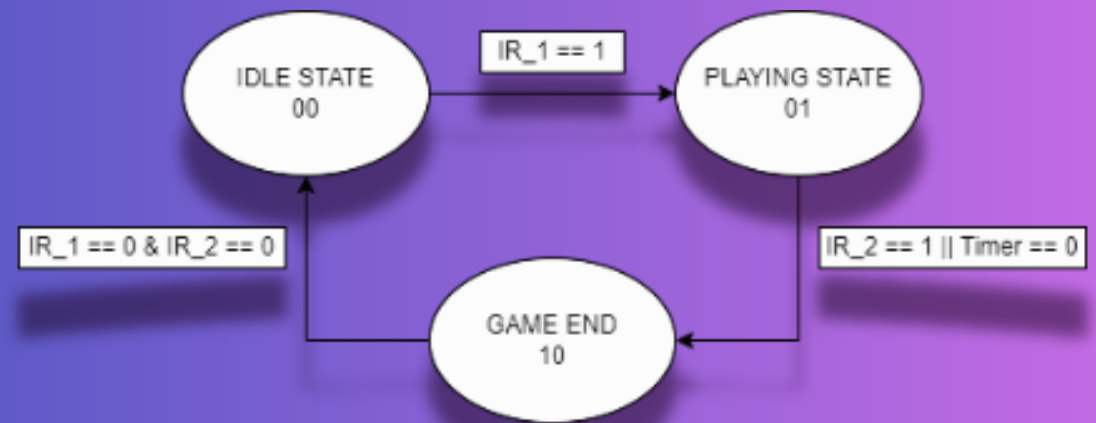
DLD PROJECT EXHIBITION AND COMPETITION (DPEC) 2024



HARDWARE RESOURCE UTILIZATION



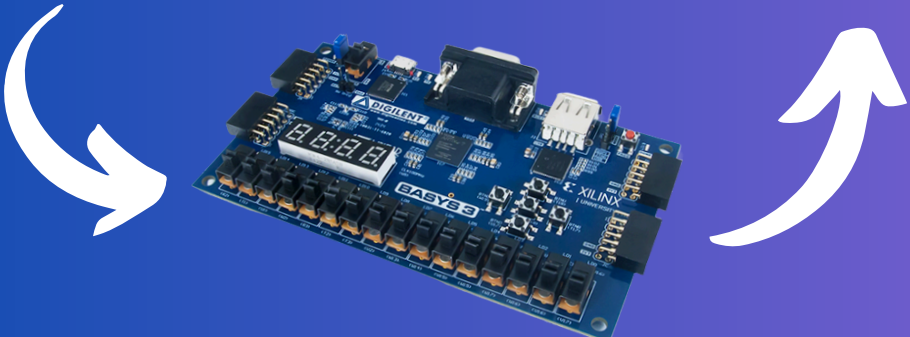
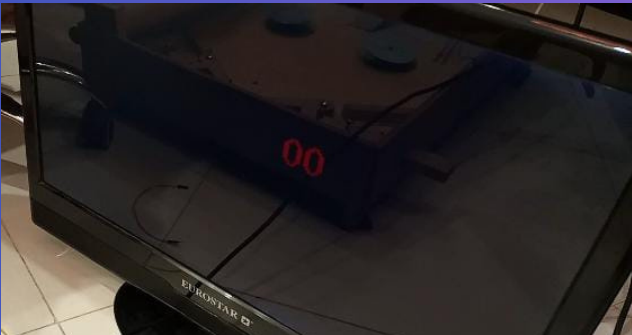
FSM STATE TRANSITION DIAGRAM



INTRODUCTION

Our project merges hardware and software seamlessly to create an engaging pinball experience. Constructed with wood and chipboard, our physical pinball arena integrates metal sheet obstacles connected to an FPGA system for real-time score tracking. IR sensors at the ball launcher and goal facilitate game initiation and completion. FPGA technology also powers digital features like a game screen displaying score, timer, and end game status. Notably, our design follows a Mealy model, where next state equations dynamically depend on inputs, enhancing gameplay dynamics.

SYSTEM OVERVIEW



STATE 00: IDLE



STATE 01: PLAYING



STATE 02: END

